

Multilingual intent router: a fine-tuned classifier with open-set abstention

Data: a synthetic logistics intent dataset (500 rows, 12 classes) provided for an assessment; not redistributed, no message text appears in this document. Every number is rendered by code from `results/*.json` (key-to-file map: `report/provenance.json` in the public repository). Labels: `MEASURED` = measured (seed, split and interval stated) · `ESTIMATE` = estimate (assumption stated) · `OPTIMISTIC` = post-selection, optimistic.

Links. Model (Hugging Face): <https://huggingface.co/gauravgandhi2411/multilingual-intent-router> · W&B report: <https://wandb.ai/gauravgandhi429-gaurav-gandhi/multilingual-intent-router/reports/Submission-summary--VmldzoxODA1MTA3Mw==> · GitHub: <https://github.com/gaurav-gandhi-2411/multilingual-intent-router> · Colab notebook: https://colab.research.google.com/github/gaurav-gandhi-2411/multilingual-intent-router/blob/v1.0-submission/notebooks/intent_router_colab.ipynb

Summary. Method: open-set behaviour is evaluated class-held-out (whole intents are removed from training), splits are stratified and duplicate-grouped, decision rules were fixed before the results, reported numbers are rendered from saved results and recomputed by a separate check, training is bitwise reproducible within a GPU type, and a Colab T4 reproduction passes (REPRODUCTION: PASS). **Results:** a fine-tuned multilingual-e5-base classifier routes 12 intents with test macro-F1 0.947 [0.865, 0.989] (`MEASURED`) one test evaluation, n = 74; floors B0 0.733 [0.602, 0.822], B1 0.899 [0.802, 0.956]) and out-of-fold macro-F1 0.953 [0.932, 0.971] (`OPTIMISTIC`). On the headline holdout at 95% known-class retention (`MEASURED`), the shipped Mahalanobis scorer rejects 0.342 [0.263, 0.425] of unknown-intent rows (strict) at 0.968 [0.925, 1.000] retention and AUROC 0.870 [0.811, 0.923], against 0.196 ± 0.104 at 0.984 ± 0.016 and AUROC 0.841 ± 0.019 for the MSP baseline. **Limit and next step:** open-set rejection is the measured limit; an oracle probe reaches AUROC 0.969 [0.942, 0.990] against 0.870 unpurvised, so the classes are separable and the missing piece is a negative training signal (an upper bound, not achievable rejection; Appendix C), and the next step is verified hard negatives and more labels. v1 ships although the selection rule, fixed before the candidates were run, picked v3, because v3 changed headline strict rejection by -0.133 [-0.204, -0.067] (paired CI; Appendix D).

Headline (<code>MEASURED</code> ; 95% intervals)	MSP baseline	Shipped scorer (v1)
Track A: test macro-F1 (primary; n = 74, one evaluation)	—	0.947 [0.865, 0.989]
Track A: test accuracy	—	0.946 [0.892, 0.986]
Track B: AUROC (headline holdout)	0.841 ± 0.019	0.870 [0.811, 0.923]
Track B: strict rejection of unknown intents at the 95% retention point	0.196 ± 0.104	0.342 [0.263, 0.425]
Track B: known retention at the same point	0.984 ± 0.016	0.968 [0.925, 1.000]

Track A: 95% percentile bootstrap (10000 resamples, seed 42), model seed 42. Track B: 3-seed means; MSP as mean ± std over seeds, the shipped Mahalanobis scorer with a 95% interval (rows resampled within the holdout, 10000 resamples).

1. Approach and Model Selection

Approach. Two parts for a 12-way, multilingual, open-set task. A closed-set classifier: multilingual-e5-base fine-tuned end to end (learning rate 3.0e-05, batch 16, stopped at epoch 9, seed 42) with temperature scaling; floors are B0 (character TF-IDF + LR) and B1 (frozen e5 + LR). And a post-hoc Mahalanobis open-set score on its penultimate features, thresholded on known-class validation rows only so that 95.0% of known rows are kept; messages below it go to an unknown path instead of a handler.

Candidate (all <code>AutoModelForSequenceClassification</code>)	CV macro-F1 (mean ± std)	params (M)	s / run
F1: multilingual-e5-base (winner)	0.939 ± 0.022	278	87.6
F2: xlm-roberta-base	0.914 ± 0.033	278	101.5
F3: mdeberta-v3-base	0.900 ± 0.029	279	167.6

`MEASURED` `OPTIMISTIC` CV on train+val only, 45 fold-runs for the best learning rate per model; test untouched. Full table: Appendix I.

Why e5-base. The selection rules were fixed before the candidates were run: highest mean CV macro-F1, and a smaller or faster model within one std wins instead. e5-base has the highest CV macro-F1 (0.939 ± 0.022); the others fall below its mean minus one std (0.917) and no rival is smaller or faster, so the second rule does not trigger (best rival 0.025 behind). Cost of the choice: the fp32 weights are 1112 MB and the shipped ONNX fp32 backend serves a batch-1 message on one CPU thread at p50 71 ms (measured on Colab).

Encoder versus LLM. Local 8B instruction models (qwen3:8b and llama3.1:8b, zero-shot and 5-shot) on the same 74 test rows reach macro-F1 0.601 to 0.680 against 0.947 [0.865, 0.989] and almost never abstain (strict rejection at most 0.013 against 0.342 ± 0.104 at 0.968 ± 0.000 retention for the shipped scorer); `MEASURED` on the same local GPU they take p50 170 to 185 ms per message against 17 ms for the encoder (at least 9.9×) and cost \$0.0166 to \$0.0179 per 1k messages against \$0.0017 (at least 9.9×; `ESTIMATE` optimistic lower bound at \$0.35 per hour). This GPU basis is the only like-for-like one: the encoder's CPU latency was measured on shared Colab vCPUs and no LLM was timed on CPU, so no CPU-versus-GPU ratio is formed (Appendix F).

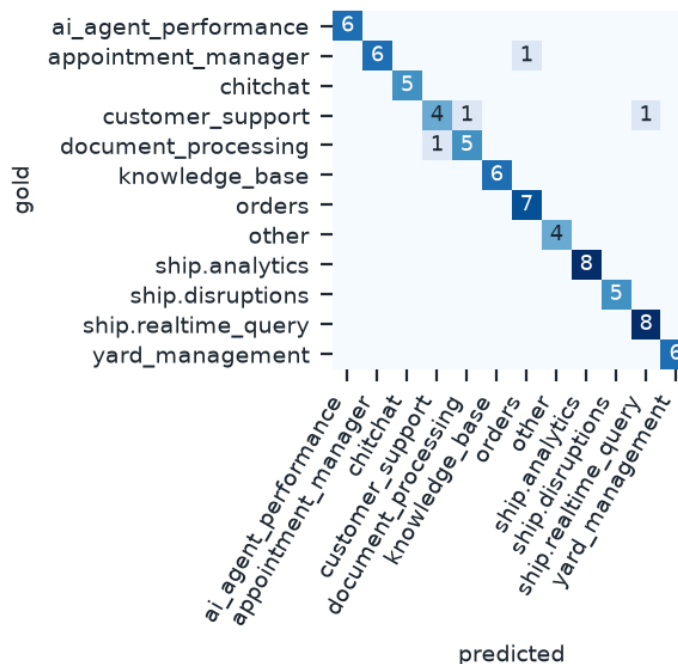
2. Data Characteristics and Handling

- Class imbalance.** Ratio 1.83 (largest class 55 rows, smallest 30): macro-F1 and stratified, duplicate-grouped splits; class-weighted cross-entropy was ablated, not adopted (CV delta -0.0027, p = 0.715).
- Multilingual.** Language ID (`MEASURED`) finds 12.4% non-English by primary language, not the stated ~25.0%; Spanish, French, German and Chinese slices are small (22, 18, 12, 7 rows), so each carries n and an interval.
- Sibling classes.** The three `shipment_information.*` classes overlap lexically (nearest-neighbour accuracy 0.594); a hierarchical auxiliary loss gave no gain over the CV noise band (delta -0.0047, p = 0.627); plain cross-entropy kept.
- Noise.** 6 rows are very short (emoji, one-liners) and kept; 52.8% of rows are noisy (33.7% all-lower-case, 6.8% slang); robustness is measured in Appendix E.
- Catch-all classes.** `chitchat` (35 rows) and `other` (30 rows) are the smallest classes (test F1 1.000, n = 5; 1.000, n = 4: small n, indicative).
- Small data.** 500 rows in total and a 74-row test split, so one error moves a class F1 a lot; weight decay 0.01 and the stopping epoch were chosen on CV, never on test, and headline numbers carry an interval.
- ID-prefix shortcut.** PO-style IDs occur in 23 rows, 82.6% labelled orders; LD-style IDs in 44 rows, 72.7% `shipment_information realtime_query`. Swapping the LD prefix for the neutral REF prefix flips 23.7% of LD rows (Appendix E).

3. Closed-Set Classification (Track A)

Why macro-F1. Each intent routes to a different handler, so the 12 classes count equally; accuracy (0.946 [0.892, 0.986]) is the sanity number and hides per-class differences at 4 to 8 test rows per class (lowest-scoring class `customer_support`, F1 0.727, support 6).

Class (test, single evaluation)	P	R	F1	n
ai_agent_performance	1.000	1.000	1.000	6
appointment_manager	1.000	0.857	0.923	7
chitchat	1.000	1.000	1.000	5
customer_support	0.800	0.667	0.727	6
document_processing	0.833	0.833	0.833	6
knowledge_base	1.000	1.000	1.000	6
orders	0.875	1.000	0.933	7
other	1.000	1.000	1.000	4
shipment_information.analytics	1.000	1.000	1.000	8
shipment_information.disruptions	1.000	1.000	1.000	5
shipment_information.realtime_query	0.889	1.000	0.941	8
yard_management	1.000	1.000	1.000	6



MEASURED confusion counts, test, n = 74.

Paired comparison on the same test rows. Fine-tuned minus B0: +0.213 [+0.113, +0.340] (McNemar exact p = 7.3e-04); minus B1: +0.048 [-0.009, +0.119] (McNemar exact p = 0.219, so the gap to B1 sits inside its interval at n = 74). Out-of-fold scores, calibration and slices: Appendix I.

4. Open-Set Abstention (Track B)

Whole classes are held out (document_processing + yard_management, headline) and removed from training and calibration; the evaluation set is 62 known test rows plus 80 unknown rows, over 3 model seeds. **Threshold rule:** the score threshold is set on known-only validation rows so that 95% of them are kept; for the shipped Mahalanobis scorer it is -128.06 (0.959 achieved on validation). Strict mode counts only unknown as a rejection.

Headline holdout (MEASURED ; 3-seed mean ± std)	AUROC	rejection @ 95%	retention @ 95%	rejection @ 90%	retention @ 90%
MSP baseline	0.841 ± 0.019	0.196 ± 0.104	0.984 ± 0.016	0.396 ± 0.019	0.941 ± 0.009
Shipped: Mahalanobis on fine-tuned features (v1)	0.870 ± 0.008	0.342 ± 0.104	0.968 ± 0.000	0.454 ± 0.036	0.946 ± 0.025

Rejection = strict rejection recall of unknown rows; retention = share of known rows kept. The 90% columns are recomputed from the saved score tables with the same threshold rule (the 95% columns are the stored values, which the recomputation reproduces). Shipped v1 with 95% intervals: rejection 0.342 [0.263, 0.425] at retention 0.968 [0.925, 1.000]; at the 90% point 0.454 [0.362, 0.546] at 0.946 [0.892, 0.989]. The full scorer table is in Appendix B.

Root cause, in one line. The oracle probe (D1) trained on the held-out-class labels reaches AUROC 0.969 [0.942, 0.990] on the unseen-known pool against 0.870 for the unsupervised scorer: the classes are separable and the missing piece is a negative training signal (an upper bound, not achievable rejection). Next step: verified hard negatives; details in Appendix C.

5. Error Analysis

Top out-of-fold confusions (45 fold-runs; **MEASURED** **OPTIMISTIC**): document_processing → knowledge_base (27); knowledge_base → document_processing (18); customer_support → appointment_manager (14). Hypotheses (next step: targeted test): document_processing and knowledge_base share document vocabulary (lexical overlap); customer_support catches complaints that mention scheduling or yard events (missing context).

Hypothesis: errors concentrate in shipment_information.* (rejected). The family is 29.1% of rows but 16.9% of errors (cluster-bootstrap interval of the difference [-0.224, -0.004]); realtime_query and disruptions are confused 7 and 0 times, 3.0% of all errors (test: 0 of 4). Tables and the ID-swap counterfactual: Appendix I and E.

6. Deployment and Monitoring

Serving. predict(text) returns label, confidence, OOD score, abstained flag and top three; below the Mahalanobis threshold the router sends the message to the unknown path instead of a handler. Default backend: ONNX fp32 on CPU, one thread, batch-1: p50 71 ms and p95 93 ms (**MEASURED** Colab shared vCPUs, environment in Appendix G; PyTorch fp32 p50 135 ms, p95 201 ms). int8 quantization did not meet the pre-set rule (macro-F1 delta at least -0.005 on both splits; validation -0.024), so fp32 ships.

Monitoring and drift. Abstention rate (test reference: 0.986 coverage), confidence and energy distributions, language mix against the measured 12.4% non-English, kNN distance to train (PSI or MMD against a reference window), per-intent volume and handler failure rate; weekly sampled label audits by humans, because consensus labelling by LLM judges did not meet its calibration gate (judge accuracy 0.70 to 0.80 against 0.90).

13th-class playbook. Cluster the abstained traffic, label the cluster, add the class at once through the kNN bank (no retraining), then retrain fully, shadow-evaluate against v1 and promote; retrain on a sustained rise in abstention, a kNN-distance PSI alert, a handler failure spike or a new labelled cluster of meaningful size.

Known Limitations, Mitigations and Next Steps

Limitation, with its measured number	Evidence	Mitigation in production today	Next step
Open-set rejection. MEASURED Headline holdout: the shipped Mahalanobis scorer rejects 0.342 of unseen-intent rows (strict) at 0.968 known retention (95% calibration target). CONFIRM classes (5, mean): strict rejection 0.227.	MEASURED The oracle probe (D1): a probe trained on the held-out-class labels reaches AUROC 0.969 [0.942, 0.990] on the unseen-known pool, against 0.870 unsupervised. The oracle is an upper bound on separability, not achievable rejection; the gap points at a missing negative training signal.	Choose the operating point by cost, using the business table in Appendix B (ESTIMATE per 1000 messages at 5% novel traffic: at the 90% point 26.8 unknowns misrouted and 65.3 known wrongly abstained; at the 95% point 38.6 and 28.3). Abstentions go to a clarifying question or a human handoff, never to a handler. Per-handler confidence gates for high-risk tools. Shadow mode records abstained and low-margin traffic.	Verified hard negatives as a 13th-class head. More labels: MEASURED the learning curve (D4): the AUROC slope is +0.056 [+0.049, +0.061] per doubling of the training data.
Fine-tune versus frozen B1. MEASURED Test, paired: macro-F1 +0.048 [-0.009, +0.119], McNemar exact p = 0.219 (5 against 1 discordant). Cross-validation: mean difference +0.025, Nadeau-Bengio p = 0.025 (15 fold-runs; a context test against the floor, not part of the selection rule, which tested the top two models).	MEASURED At 74 test rows the paired interval is 0.129 wide and includes zero (test macro-F1 interval width 0.124).	Ship the fine-tune: it has the better CV macro-F1 (0.939 against 0.914 for B1; OPTIMISTIC post-selection). Track B (report-only): B1 with MSP scores DEV AUROC 0.836 against 0.894 for the shipped Mahalanobis, and CONFIRM 0.856 against 0.879 (headline holdout, shipped: 0.870 ± 0.008). Keep B1 as a training-free fallback: same inference cost (same frozen e5 encoder) and a logistic-regression head to retrain, in seconds (e.g. for a new intent).	A larger labelled test set to resolve the observed accuracy difference (McNemar). ESTIMATE (observed discordance): the McNemar sample-size approximation $n = (z(1-\alpha/2) \cdot \sqrt{\psi} + z(1-\beta) \cdot \sqrt{(\psi - \delta^2)})^2 / \delta^2$, with $\psi = (b + c) / n_{\text{test}}$ and $\delta = (b - c) / n_{\text{test}}$ = 0.054 from the observed discordant counts (b = 5, c = 1), 0.05 two-sided and 80% power, gives about 216 rows (2.9× the current 74). It assumes the observed discordance rate (0.081 of rows) holds, which is itself uncertain.
ID-prefix shortcut. MEASURED OPTIMISTIC Swapping the LD prefix for the neutral REF prefix flips 23.7% of 38 LD-prefixed out-of-fold predictions (9 flips).	MEASURED EDA shortcut audit: 72.7% of the 44 LD-prefixed rows carry shipment <code>information realtime query</code> (strong, not deterministic); 82.6% of 23 PO-prefixed rows carry <code>orders</code> (flagged as a shortcut risk: yes).	Normalise IDs before routing, or use the v3 variant in which IDs vary (not shipped: it moved headline strict rejection by -0.133 [-0.204, -0.067]). Monitor the flip rate on production traffic.	A consistency-loss variant that keeps Track A. Measured trade-off: neutral-swap flip rate -0.183 and -0.198 (two loss weights) against CV macro-F1 -0.023 [-0.033, -0.013] and -0.023 [-0.032, -0.015], so it needs a variant that recovers that loss.
Small synthetic data. MEASURED 500 rows from one synthetic source; the test split has 74 rows (4 to 8 per class).	MEASURED Test macro-F1 interval width 0.124 (0.947 [0.865, 0.989]): one error moves a class F1 a lot at this n.	The numbers are provisional. Re-baseline on a labelled sample of real traffic before trusting any threshold.	Active learning from production traffic: label abstained and low-margin messages first, and grow the test set with them.

7. Next Steps

- **Verified hard negatives (first step).** Held-out intents are separable (oracle AUROC 0.969 [0.942, 0.990]) but the unsupervised score reaches 0.870: generate out-of-scope messages that reuse keywords of the 12 intents, keep those all three judges unanimously label as fitting none, and compare a K-plus-one class head against outlier exposure on the same Track B holdouts.
- **More labelled data.** The measured learning curve (D4): the AUROC slope is +0.056 [+0.049, +0.061] per doubling of the training data, but the rejection slope (+0.061 [-0.010, +0.145]) has an interval reaching zero, so label abstained and low-margin traffic first.
- **An ID-consistency variant with its measured trade-off.** The consistency loss changed the neutral-swap flip rate by -0.183 and -0.198 against CV macro-F1 -0.023 [-0.033, -0.013] and -0.023 [-0.032, -0.015] (two loss weights); it needs a variant that recovers the F1 loss and must be guarded on Track B too.

Appendix contents. A Debrief Answers · B Open-Set Evaluation in Full: Scorers, Leave-One-Class-Out and Business Framing · C Diagnostics (Oracle Probe, ID-Neutral Scoring, Threshold Stability, Learning Curve) · D Selection History and the v1/v3 Decision · E Robustness · F LLM Baseline · G Latency, Quantization and Cost · H Reproduction and Determinism · I Ablations and Bake-Off Detail · J Test-Split Exposure · K Further Detail

Candidate codes used in this appendix (each defined once). A1 = random ID prefixes in training text; A2 = outlier exposure with generic synthetic out-of-scope text; A3 = machine-translated and noisy copies of training rows. I1 = alternative open-set scorers: i1a = relative Mahalanobis (class distance minus a class-agnostic background distance), i1b = L2-normalised Mahalanobis on the same weights. I4 = an ID-consistency loss (a symmetric KL between a row and its prefix-swapped twin), i4a and i4b being two loss weights. I6 = alternative thresholding: i6a = calibration rows grouped by predicted class, i6b = cross-fitted Gaussian. N1 = verified hard negatives as training signal; N2 = a fine-tuned e5-large. D1 = the oracle probe, D2 = ID-neutral Track B scoring, D3 = threshold stability, D4 = the data-scaling slope.

A. Debrief Answers

Debrief answers. One short paragraph for each question we expect in the walk-through.

Why this metric, and what does accuracy hide? Every intent routes to a different handler, so each of the 12 classes must count equally: macro-F1 is the primary metric, accuracy the sanity number. On this mildly imbalanced data (largest class 55 rows, smallest 30, ratio 1.83) the two are close on test (0.947 [0.865, 0.989] vs 0.946 [0.892, 0.986]), so accuracy hides little in aggregate but does hide per-class differences at 4 to 8 test rows per class. The lowest-scoring test class is customer_support (F1 0.727, support 6), noise-dominated at that n, hence the CV and CIs below.

With 500 rows, how do we trust the test number? Not alone: its interval is wide (0.947 [0.865, 0.989]), so three estimates are reported with their roles. The single test split is the unbiased one; the 45-run CV (0.939 ± 0.022) and the out-of-fold score (0.953 [0.932, 0.971]) are **OPTIMISTIC**, as the model and epoch were selected on them. Against baselines the fine-tune clearly beats B0 (paired macro-F1 difference +0.213 [+0.113, +0.340]) but not provably B1 on test (+0.048 [-0.009, +0.119], the interval includes zero; exact McNemar p = 0.219); in CV it beats B1 (Nadeau-Bengio p = 0.025).

The model confuses realtime_query and disruptions: what now? In the 45 out-of-fold fold-runs these two siblings are confused 7 times (realtime as disruptions) and 0 times (the reverse), 3.0% of all errors; on the test split 0 of 4 errors. The hypothesis that errors concentrate in the shipment family is rejected: the family is 29.1% of rows but 16.9% of errors; real confusion sits in document_processing vs knowledge_base and customer_support (see section 5). Lexical overlap in the family is real (nearest-neighbour accuracy 0.594), but a hierarchical auxiliary loss gave no gain over the CV noise band (delta -0.0047, p = 0.627); plain cross-entropy kept. Next: temporal-cue labelling rules and more disruption examples, not more modelling.

A brand-new intent appears next quarter. What happens, and how do we add a 13th class? The router abstains on 0.342 [0.263, 0.425] of unseen-intent messages at 0.968 [0.925, 1.000] known retention, so about 65.8% of novel intents (headline holdout) are still routed to a known label (most often shipment_information.realtime_query). Per 1000 messages at 5% novel-intent traffic and the 95% operating point (**ESTIMATE** using the CONFIRM-mean strict rejection 0.227, not the headline rate above): 11.4 caught, 38.6 misrouted, 28.3 known messages wrongly abstained. The extension path (cluster abstained traffic, label, kNN bank, retrain, shadow, promote) is in section 6.

Why is softmax accuracy zero on a class it never saw, and how did we measure open-set behaviour? It can only emit one of its trained labels, so accuracy on a held-out class is zero by construction. We therefore hold out whole classes (headline: document_processing + yard_management; plus 10 leave-one-class-out runs), set the threshold on known-only validation data at fixed retention, and report rejection recall, retention and AUROC. MSP reaches AUROC 0.841 ± 0.019 on the headline holdout; the shipped Mahalanobis scorer 0.870 ± 0.008.

Does the metric hold on the non-English slice? The stated share of about 25.0% non-English is not what the data show: language ID finds 12.4%. On the 49 non-English out-of-fold rows macro-F1 is 0.963 [0.821, 1.000] (wide interval); the test split has 13 rows, indicative. Under machine translation of English rows accuracy drops from 0.924 to 0.778 for Chinese, the worst language (**MEASURED**, MT evaluation data, see Appendix E).

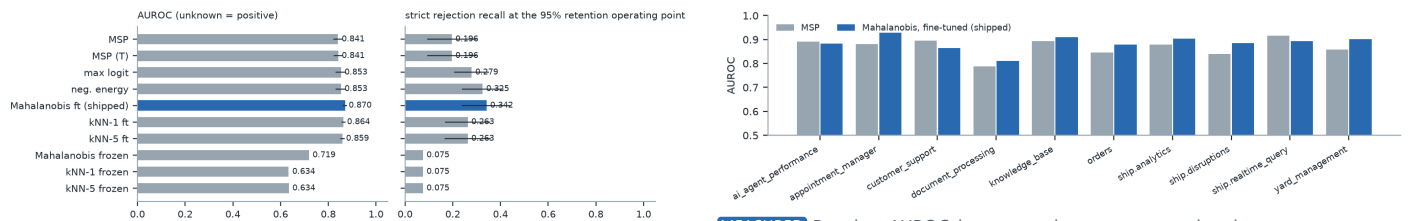
How do we monitor and catch data and label drift? Abstention rate (test reference: 73 of 74 rows accepted), confidence and energy distributions, language mix, kNN distance to train; label audits by multi-model consensus did not meet our own calibration gate (judge accuracy 0.70 to 0.80 against 0.90), so humans label the sample. Detail in section 6 and Appendix K.

B. Open-Set Evaluation in Full: Scorers, Leave-One-Class-Out and Business Framing

Holdout design: document_processing + yard_management are removed from training and calibration; the evaluation set is 62 known test rows plus 80 unknown rows (all rows of the held-out classes, since the model never saw them). 3 model seeds (42, 43, 44); closed-set macro-F1 on the known side 0.960 ± 0.008. Threshold: 95% retention on known-only validation rows. Strict mode counts only unknown as a rejection.

Score (higher = known)	AUROC	strict rejection	lenient rejection	retention	LOCO AUROC	LOCO rejection
MSP (softmax)	0.841 ± 0.019	0.196 ± 0.104	0.196 ± 0.104	0.984 ± 0.016	0.869	0.301
MSP, temperature-scaled	0.841 ± 0.019	0.196 ± 0.104	0.196 ± 0.104	0.984 ± 0.016	0.869	0.305
max logit	0.853 ± 0.022	0.279 ± 0.073	0.279 ± 0.073	0.973 ± 0.009	0.877	0.347
negative energy	0.853 ± 0.023	0.325 ± 0.087	0.325 ± 0.087	0.968 ± 0.016	0.880	0.380
Mahalanobis, fine-tuned features (shipped)	0.870 ± 0.008	0.342 ± 0.104	0.342 ± 0.104	0.968 ± 0.000	0.886	0.330
kNN k=1, fine-tuned features	0.864 ± 0.009	0.263 ± 0.094	0.263 ± 0.094	0.984 ± 0.016	0.884	0.370
kNN k=5, fine-tuned features	0.859 ± 0.010	0.263 ± 0.098	0.263 ± 0.098	0.984 ± 0.016	0.882	0.355
Mahalanobis, frozen e5 features	0.719 ± 0.000	0.075 ± 0.000	0.075 ± 0.000	0.935 ± 0.000	0.676	0.062
kNN k=1, frozen e5 features	0.634 ± 0.000	0.075 ± 0.000	0.075 ± 0.000	0.903 ± 0.000	0.662	0.092
kNN k=5, frozen e5 features	0.634 ± 0.000	0.075 ± 0.000	0.075 ± 0.000	0.903 ± 0.000	0.637	0.060

MEASURED headline columns: mean ± std (sample) over 3 model seeds; LOCO columns: mean over 10 runs, one seed each. Shipped score chosen by best mean LOCO AUROC (a design choice made across holdout experiments; kNN with one neighbour is a near tie). Lenient mode (other and chitchat count as safe) equals strict: yes.



Headline holdout, mean ± std over seeds.

MEASURED Per-class AUROC, leave-one-class-out, one seed each.

Shipped v1, with intervals. Headline: AUROC 0.870 [0.811, 0.923], strict rejection 0.342 [0.263, 0.425] at retention 0.968 [0.925, 1.000]; at the 90% operating point rejection 0.454 [0.362, 0.546] at retention 0.946 [0.892, 0.989] (**MEASURED** rows resampled within the holdout, stratified by known/unknown, 10000 re-samples). CONFIRM classes (5 classes, not used for selection): AUROC 0.879 [0.850, 0.908], rejection 0.227 [0.174, 0.283]. DEV classes (5, used for selection): AUROC 0.894, rejection 0.432.

Reading. Fine-tuned features beat frozen ones by a wide margin for distance scores; MSP, energy and Mahalanobis are close. Un-rejected unknowns mostly land on shipment_information.realtime_query (29.7%), ai_agent_performance (20.9%) and knowledge_base (14.6%); 0 of 158 reach a catch-all class. Label smoothing changed headline AUROC by -0.005 ± 0.005.

ESTIMATE Business framing per 1000 messages. Assumptions: novel-intent prevalence as listed; rates are the measured CONFIRM-mean strict rejection (0.227 at the high and 0.464 at the low operating point) and retention.				
Calibration retention	novel prevalence	known wrongly abstained	novel caught	novel misrouted
90%	1%	68.0	4.6	5.4
90%	5%	65.3	23.2	26.8
90%	10%	61.8	46.4	53.6
95%	1%	29.5	2.3	7.7
95%	5%	28.3	11.4	38.6
95%	10%	26.8	22.7	77.3

Basis: the per-message rates here use the CONFIRM-mean rejection rate, while the share of unknowns routed to a known label in the text (65.8%) is one minus the headline-holdout rate. Formulas: wrongly abstained = $N(1-\pi)(1-\text{retention_measured})$; caught = $N\pi\text{rejection_recall}$; misrouted = $N\pi(1-\text{rejection_recall})$.

C. Diagnostics (Oracle Probe, ID-Neutral Scoring, Threshold Stability, Learning Curve)

Measured gap. **MEASURED** On the headline holdout (document_processing + yard_management) the shipped scorer has AUROC 0.870 [0.811, 0.923] but rejects 0.342 [0.263, 0.425] of unknown-intent rows at 0.968 [0.925, 1.000] known retention (calibration target 95%); on the 5 CONFIRM classes rejection is 0.227 [0.174, 0.283]. 65.8% of unknowns are therefore routed to a known label, and 0 of the 158 un-rejected unknown rows land in a catch-all class. At the 95% operating point and 5% novel-intent traffic per 1000 messages that is 38.6 misrouted, computed with the CONFIRM-mean strict rejection 0.227 rather than the headline-holdout rate quoted above (**ESTIMATE**).

Root cause: facts from the Open-set improvement round diagnostics (report-only, never selection inputs; tables below).

- D1, ORACLE separability.** Held-out intents are linearly separable in the fine-tuned feature space (oracle AUROC 0.969 [0.942, 0.990] on the unseen-known pool), yet unsupervised scoring reaches 0.870; the gap, +0.098, is a missing negative training signal, which verified hard-negative OOS data (next step) supplies. The probe is trained on the held-out-class labels (headline holdout; pool = 80 unknown and 124 known rows the encoder never trained on), so the oracle AUROC is an upper bound on separability, not achievable rejection. Pools: the oracle figure drops the known training rows (the v1 encoder trained on them, which inflates the all-rows figure; that figure is in the D1 table below, marked as inflated); the unsupervised figure is v1's seed-mean Mahalanobis AUROC on the Track B evaluation rows, which the encoder also never trained on.
- D2, ID-neutral Track B** (IDs rewritten at scoring time; headline). v1 strict rejection 0.342 [0.263, 0.425] raw, 0.292 [0.217, 0.371] neutral. v3 minus v1 in rejection: -0.133 [-0.204, -0.067] raw, -0.063 [-0.129, +0.004] neutral; ranking reversal: no. v1's Track B lead over v3 therefore holds in direction under ID-neutral scoring, so the ranking is not an ID-shortcut artifact; the neutral lead is smaller and its interval reaches zero, so part of the raw gap is ID text and the neutral lead is a direction, not a significant gap. The v1 decision stands.
- D3, threshold stability** (2000 resamples of the known calibration rows, 3 seeds pooled). Retention sd 0.019 (95% range 0.919 to 0.984); unknown rejection sd 0.108 (95% range 0.175 to 0.550). On the evaluation rows the cross-fitted threshold gives retention 0.371 and rejection 0.967 (validation threshold: table below): it collapses known retention, so thresholds must be calibrated on the shipped model's own held-out data.
- D4, learning curve** (77 to 296 training rows, 25% to 100% of the data; rows below). **MEASURED** Slope per doubling of the training data (CI over 3 seeds, coarse): AUROC +0.056 [+0.049, +0.061], strict rejection +0.061 [-0.010, +0.145]. **ESTIMATE beyond 296 rows**, assuming AUROC stays log-linear in rows (fit 0.876 at 296 rows plus the slope per doubling; interval from the slope CI only): 592 rows (2×) 0.932 [0.925, 0.937]. The line would hit the AUROC ceiling after 2.2 doublings, so the real curve must bend before then; the seed CI is coarse and no data beyond 296 rows exists.

D1 ORACLE probe features	AUROC, unseen known only (headline)	AUROC, all rows (inflated for fine-tuned)	D2, headline	AUROC raw	AUROC neutral	strict rej. raw	strict rej. neutral
fine-tuned v1 features	0.969 [0.942, 0.990]	0.993 [0.987, 0.997]	v1 (shipped)	0.870 [0.811, 0.923]	0.842 [0.777, 0.900]	0.342 [0.263, 0.425]	0.292 [0.217, 0.371]
frozen e5-base	0.952 [0.919, 0.979]	0.970 [0.943, 0.990]	v3	0.829 [0.762, 0.890]	0.829 [0.762, 0.888]	0.208 [0.142, 0.283]	0.229 [0.163, 0.304]
frozen e5-large	0.970 [0.943, 0.990]	0.974 [0.950, 0.991]	A1	0.847 [0.783, 0.904]	0.838 [0.773, 0.897]	0.346 [0.258, 0.433]	0.354 [0.267, 0.446]
frozen bge-m3	0.975 [0.953, 0.993]	0.979 [0.963, 0.991]	v3 minus v1	-0.041 [-0.076, -0.010]	-0.013 [-0.056, +0.031]	-0.133 [-0.204, -0.067]	-0.063 [-0.129, +0.004]
frozen LaBSE	0.899 [0.848, 0.946]	0.940 [0.912, 0.961]	MEASURED Seed mean, bootstrap CI; v3 minus v1 is paired. Ranking reversal between modes: no. ID-bearing rows: 10 of 62 known evaluation, 25 of 80 unknown.				
unsupervised Mahalanobis, v1 (seed mean)	0.870						

MEASURED ORACLE: the probe is trained on the held-out-class labels, so it is an upper bound on separability, not achievable rejection. Headline holdout, mean of folds [percentile CI over near-duplicate groups]; the headline column drops the known training rows the v1 encoder saw; the all-rows column includes them, so for the fine-tuned features it is inflated.

D3, v1 headline	known retention	unknown rejection	D4, training fraction	training rows	AUROC	strict rejection
bootstrap mean	0.962	0.362	25%	77	0.758	0.217
bootstrap sd	0.019	0.108	50%	151	0.833	0.312
95% range, low	0.919	0.175	75%	226	0.853	0.317
95% range, high	0.984	0.550	100%	296	0.870	0.342
validation threshold (seed mean)	0.968	0.342	slope per doubling			+0.056 [+0.049, +0.061]
cross-fitted threshold (seed mean)	0.371	0.967				+0.061 [-0.010, +0.145]

MEASURED 2000 resamples of the known calibration rows, 3 seeds pooled; the last two rows are on the held-out evaluation rows (mean thresholds -117.2 and -38.7).

What was tried. **MEASURED** On DEV AUROC (current scorer 0.894): rank fusion of Mahalanobis and energy 0.890, multi-layer Mahalanobis 0.832, CE + SupCon 0.887 and 0.884, none cleared the margin 0.0115 (ship: no). Outlier exposure (A2) changed DEV AUROC by -0.037 [-0.056, -0.019] and DEV rejection by -0.126 [-0.198, -0.056]; label smoothing changed headline AUROC by -0.005 ± 0.005; v2 and v3 moved headline rejection by -0.117 [-0.188, -0.050] and -0.133 [-0.204, -0.067].

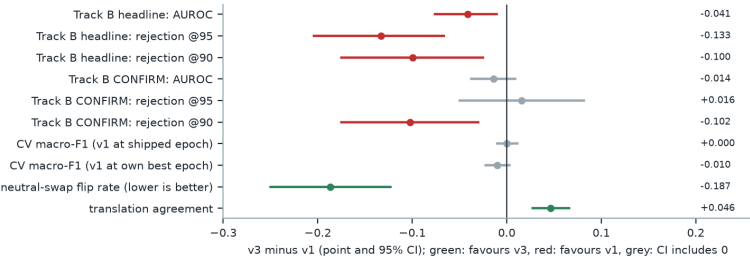
What would break it. (i) D1 is an upper bound: verified hard-negative data on classes never trained on (N1) is the test of whether the headroom of +0.098 is reachable. (ii) D2 rules out the ID shortcut as the cause of the v1-versus-v3 ranking, not of closed-set accuracy riding on ID text (Appendix E). (iii) Rejection differences below the D3 spread (sd 0.108) are noise unless both arms share the threshold rows, as in the paired bootstrap of the open-set improvement round. (iv) If the D4 rejection slope CI (+0.061 [-0.010, +0.145]) keeps reaching zero, more data of the same kind is not the lever for rejection.

D. Selection History and the v1/v3 Decision

3 final models were trained and each was test-evaluated once (MEASURED single recorded evaluation, 0.947 [0.865, 0.989] for v1, 0.949 [0.865, 0.990] for v2, 0.961 [0.879, 1.000] for v3: differences far inside the intervals). v2 (adds machine-translated and noisy copies of train rows, factor A3) was adopted under a rule that guarded Track A only, then reverted: on the scored Track B headline its strict rejection fell by -0.117 [-0.188, -0.050] (paired), neutral-swap flips rose from 0.274 to 0.369, and calibration temperature moved from 1.023 to 1.516. Multi-axis selection then selected on several axes with several seeds; the selection rule, fixed before the candidates were run, picked v3 (A1+A3, epoch 6). **Decision: v1 ships, overriding the rule's pick.** Reasons, in order: (a) the scored Track B headline regressed for v3 (table below); (b) v3's eligibility depended on v1 being read at its shipped epoch 9, a local dip (own best epoch 8; CV F1 0.9336 vs 0.9440), and against the best epoch the outcome flips (same outcome: no); (c) the DEV axis was not predictive of the headline regression. Test macro-F1 played no part.

v3 minus v1 (v1 is shipped)	v1	v3	delta [95% CI]
CV macro-F1, v1 at shipped epoch	0.934	0.934	+0.000 [-0.010, +0.011]
CV macro-F1, v1 at its own best epoch	0.944	0.934	-0.010 [-0.023, +0.003]
Track B DEV AUROC	0.880	0.878	-0.002 [-0.018, +0.013]
Track B DEV strict rejection @95	0.420	0.402	-0.019 [-0.067, +0.031]
Track B HEADLINE AUROC	0.870 [0.811, 0.923]	0.829 [0.762, 0.890]	-0.041 [-0.076, -0.010]
Track B HEADLINE strict rejection @95	0.342 [0.263, 0.425]	0.208 [0.142, 0.283]	-0.133 [-0.204, -0.067]
Track B HEADLINE strict rejection @90	0.454 [0.362, 0.546]	0.354 [0.275, 0.438]	-0.100 [-0.175, -0.025]
Track B CONFIRM AUROC	0.879 [0.850, 0.908]	0.865 [0.834, 0.895]	-0.014 [-0.037, +0.009]
Track B CONFIRM strict rejection @95	0.227 [0.174, 0.283]	0.243 [0.195, 0.296]	+0.016 [-0.050, +0.081]
Track B CONFIRM strict rejection @90	0.464 [0.402, 0.527]	0.361 [0.303, 0.421]	-0.102 [-0.175, -0.030]
neutral-swap flip rate (lower is better)	0.238	0.052	-0.187 [-0.250, -0.123]
translation agreement	0.853	0.900	+0.046 [+0.027, +0.066]
OOE ECE (lower is better)	0.026	0.026	-0.000 [-0.012, +0.011]

MEASURED Source: results/tradeoff_v1_v3.json (built from results JSON only). Guidance recorded there: v1 = ship default (better open-set rejection on the scored headline holdout); v3 = choose when ID-prefix shortcut robustness and translation robustness matter more than open-set rejection. Lessons: guard every scored axis at the operating point; take the reference at its own best epoch.



MEASURED v3 minus v1 with paired-bootstrap intervals (Track B holdouts, CV, robustness).

Open-set improvement round. The selection rule fixed before the runs picked i6b, a collapsed operating point (reference retention 0.968). A retention check was added after the first selection picked a candidate that kept 0.456 of known messages (0.340 under ID-neutral scoring); it then picked i1b, the only eligible candidate. The pre-ship check **held back** i1b (headline strict rejection -0.033 [-0.062, -0.008], paired CI entirely below zero), so v1 ships. i4 (ID-consistency loss) changed the neutral-swap flip rate by -0.183 and -0.198 against CV macro-F1 -0.023 [-0.033, -0.013] and -0.023 [-0.032, -0.015], and the seed soups (i3) were ineligible on CV macro-F1 (-0.020 uniform, -0.025 greedy).

Candidate	(a) CV F1	(b) raw AUROC	(b) raw rej.	(b) neutral AUROC	(b) neutral rej.	retention raw	retention neutral	(c) flip	(d) agree	(e) ECE	eligible	unmet axes	improved axes
ref (v1 recipe)						0.968	0.967						
i1a	+0.000	-0.012	-0.077	-0.009	-0.046	0.957	0.961	+0.000	+0.000	+0.000	no	b	none
i1b	+0.000	+0.001	-0.002	+0.002	+0.039	0.972	0.967	+0.000	+0.000	+0.000	yes	none	b
i6a	+0.000	+0.000	+0.062	+0.000	+0.133	0.878	0.876	+0.000	+0.000	+0.000	no	b	b
i6b	+0.000	+0.000	+0.554	+0.000	+0.691	0.456	0.340	+0.000	+0.000	+0.000	no	b	b
i3u	-0.020	-0.019	-0.087	-0.018	-0.017	0.948	0.939	+0.016	-0.004	+0.007	no	a,b	none
i3g	-0.025	-0.021	-0.085	-0.020	+0.002	0.949	0.941	+0.067	-0.007	+0.000	no	a,b,c	none
i4a	-0.023	-0.019	-0.061	-0.000	+0.062	0.965	0.964	-0.183	-0.016	-0.001	no	a,b,d	b,c
i4b	-0.023	-0.020	-0.034	+0.000	+0.097	0.957	0.956	-0.198	-0.018	+0.001	no	a,b,d	b,c
i3i1	-0.020	-0.014	-0.097	-0.011	+0.011	0.950	0.936	+0.016	-0.004	+0.007	no	a,b	none

MEASURED Candidate minus reference, point estimates on DEV (CV for (a)); the paired-bootstrap CIs are in results/phase6a/axes.json. Retention columns are the candidate's own value (guard: no more than 0.02 below the reference in both modes; added after the first selection picked a candidate that kept 0.456 (raw) and 0.340 (neutral) of known DEV rows); flip rate and ECE: negative is better. Not formed: i3i4 (no i4 variant is eligible on axis (a)); i3i4i1 (no i4 variant is eligible on axis (a)). Selected under the guard: i1b; under the rule fixed before the runs: i6b.

Pre-ship check, i1b minus ref	AUROC	strict rejection	retention
raw, headline	+0.000 [-0.003, +0.004]	-0.033 [-0.062, -0.008]	+0.005 [+0.000, +0.016]
raw, CONFIRM	+0.005 [+0.002, +0.009]	+0.017 [-0.015, +0.051]	+0.006 [+0.000, +0.015]
neutral, headline	+0.001 [-0.003, +0.004]	+0.037 [+0.008, +0.071]	+0.000 [+0.000, +0.000]
neutral, CONFIRM	+0.005 [+0.001, +0.010]	+0.010 [-0.007, +0.025]	-0.003 [-0.009, +0.000]

MEASURED Paired bootstrap deltas with CI. Rule (retention check added after the first selection): the candidate ships only if no paired CI for AUROC, strict rejection or retention lies entirely below zero on the headline holdout and on CONFIRM; one does (raw headline rejection), so it is held back and v1 with its scorer ships. The reference of the open-set improvement round is the v1 recipe at epoch 8, a different artifact from the shipped v1 at epoch 9: headline strict rejection 0.429 [0.337, 0.521] against 0.342 [0.263, 0.425] (overlapping CIs: a lead, not a finding). CONFIRM and headline looks including the open-set improvement round: 5 and 5.

E. Robustness

Language (MT rows)	n	agreement with English pred.	acc.	English acc. (same rows)
Spanish	370	0.886	0.868	0.924
French	370	0.854	0.824	0.924
German	370	0.859	0.841	0.924
Chinese	370	0.789	0.778	0.924

MEASURED **OPTIMISTIC** English out-of-fold rows translated by facebook/nllb-200-distilled-600M and labelled machine-translated (synthetic); greedy decoding (beams: 1). Synthetic MT data: an upper bound on language sensitivity (greedy decoding sometimes truncates the second clause), not native non-English performance. Test-split numbers are in the appendix tables of the results files.

ID swap (OOF rows)	n	flip rate	acc. clean → swapped
PO->LD	19	63.2%	0.895 → 0.316
PO->REF	19	36.8%	0.895 → 0.579
LD->PO	38	81.6%	0.974 → 0.184
LD->REF	38	23.7%	0.974 → 0.737
other->REF	27	25.9%	0.926 → 0.741

MEASURED **OPTIMISTIC** Counterfactual: swap the ID prefix, keep the digits; flip = prediction differs from the same model's clean prediction. PO and LD prefixes act as label cues; the neutral REF prefix still flips 23.7% of LD rows.

F. LLM Baseline

MEASURED Four local 8B configurations (qwen3:8b and llama3.1:8b; zero-shot and 5-shot) on the 74 test rows reach macro-F1 0.601 to 0.680 against 0.947 [0.865, 0.989] for the fine-tuned v1; on the headline holdout their strict rejection is at most 0.013 against 0.342 ± 0.104 for the shipped scorer. **ESTIMATE** At \$0.0166 to \$0.0179 per 1k messages (optimistic lower bound) they cost at least 9.9× the encoder's \$0.0017 (same GPU basis); the full table is below.

Why an encoder fine-tune and not an LLM. We measured the alternative: local 8B instruction models (qwen3:8b and llama3.1:8b, zero-shot and with 5 few-shot examples) on the same 74 test rows (all numbers **MEASURED**; costs **ESTIMATE**). *Quality:* LLM macro-F1 is 0.601 to 0.680 against 0.947 [0.865, 0.989]; the best LLM configuration (llama3.1:8b, zero-shot) minus v1 is -0.266 [-0.411, -0.159] (paired bootstrap). *Abstention:* the LLMs almost never answer unknown (at most 0.013 strict rejection on the headline holdout, 1.000 retention for the best) against 0.342 ± 0.104 at 0.968 ± 0.000 retention for the shipped scorer. *Latency:* p50 170 to 185 ms per message on the local GPU against 17 ms for the encoder on the same GPU (at least 9.9×). *Cost:* \$0.0166 to \$0.0179 per 1k messages on a T4-class stand-in at \$0.35 per hour (optimistic, a lower bound) against \$0.0017 for the encoder on the same basis (at least 9.9×, the GPU latency ratio, as the hourly price is shared). No CPU-versus-GPU ratio is formed: the encoder's CPU latency (ONNX fp32, one thread, p50 71 ms) was measured on shared Colab vCPUs, whereas the LLM latencies above were measured on the local NVIDIA GeForce RTX 3070 Laptop GPU, and no LLM was timed on CPU, so the GPU comparison above is the only like-for-like one. *Where the LLM has the advantage (outside these measurements):* it needs no labelled training data and takes a new intent through the prompt, while the encoder needs labels and a retrain (or the kNN-bank shortcut in section 6). Scope: local open-weight models only, no frontier API.

Local 8B model, quantized	test macro-F1 [CI]	test acc.	vs v1	strict rejection	retention	unknown answers	p50 s	p95 s	\$ / 1k
qwen3:8b, zero-shot	0.661 [0.524, 0.749]	0.676	-0.285 [-0.414, -0.193]	0.000	0.984	4.1%	0.181	0.222	\$0.0176
qwen3:8b, few-shot	0.601 [0.477, 0.685]	0.662	-0.345 [-0.465, -0.245]	0.000	1.000	0.0%	0.185	0.233	\$0.0179
llama3.1:8b, zero-shot	0.680 [0.530, 0.774]	0.703	-0.266 [-0.411, -0.159]	0.000	1.000	0.0%	0.170	0.208	\$0.0166
llama3.1:8b, few-shot	0.654 [0.521, 0.748]	0.703	-0.293 [-0.420, -0.189]	0.013	1.000	0.0%	0.170	0.215	\$0.0166
fine-tuned v1 (this report)	0.947 [0.865, 0.989]	0.946		0.342 ± 0.104	0.968 ± 0.000		0.017	0.023	\$0.0017

MEASURED Track A on the 74 test rows, 10000-resample bootstrap; Track B on the headline holdout (62 known, 80 unknown), 5 few-shot examples drawn from train. Latency is sequential on the local GPU. **ESTIMATE** cost per 1k messages = p50 latency x price per hour: LLM rows 1 T4-class GPU at \$0.35/hr; the fine-tuned row uses its GPU batch-1 latency (1 T4-class GPU at \$0.35/hr). LLM GPU rows are optimistic (a T4 is slower than the local GPU), so those costs are lower bounds.

G. Latency, Quantization and Cost

Serving path. predict(text) returns label, confidence, OOD score, abstained flag and top three; below the Mahalanobis threshold (-128.06, set at 95.0% known retention on validation, 0.959 achieved) the router sends the message to the unknown path instead of a handler. Default backend: ONNX fp32 on CPU (FastAPI endpoint, Docker image). **MEASURED** ONNX fp32 matches PyTorch: prediction agreement 1.000, max probability difference 1.4e-06; the Docker image (2.10 GB, ready in 19.2 s) reproduced 20 probe strings with equal labels: yes, max OOD-score difference 4.3e-14. Hub round trip on test: label agreement 1.000, max probability difference 1.9e-06 (tolerance 1.0e-05). A repeat training run was bitwise identical: yes.

int8 quantization: rejected. Encoder-only dynamic int8 shrinks the model to 858 MB from 1112 MB (fp32) and does not meet the pre-registered rule (macro-F1 delta at least -0.005 on both splits): validation -0.024 (pass: no), which is the deciding criterion; the test delta +0.014 (n = 74) is noise, not the criterion; abstention agreement with PyTorch 0.973, max probability difference 0.335. Decision: rejected.

Backend	test macro-F1	val macro-F1	size (MB)
PyTorch fp32	0.947	0.889	1112
ONNX fp32 (shipped)	0.947		1112
ONNX int8, encoder (rejected)	0.961	0.865	858

MEASURED Inference-only; 3 int8 variants were tried, selected on validation.

CPU batch-1 latency, ms (measured on Colab; environment stated)	one thread p50	p95	8 threads p50	p95
PyTorch fp32	135	201	202	376
ONNX fp32 (shipped default)	71	93	111	159

CPU recommendation: ONNX fp32 at one thread for batch-1 serving. **MEASURED** At one thread it has p50 71 ms and p95 93 ms against p50 135 ms and p95 201 ms for PyTorch fp32; with 8 threads ONNX fp32 has p50 111 ms and p95 159 ms, slower than one thread (PyTorch fp32: p50 202 ms, p95 376 ms, also slower). Cause not isolated; for throughput, run more single-thread replicas (concurrent load is a measurement still to take). **ESTIMATE CPU cost per 1k messages:** \$0.00098 for ONNX fp32 (\$0.00187 for PyTorch fp32), assuming sequential batch-1 calls on one vCPU at \$0.05 per hour: p50 seconds per call × 1k calls ÷ seconds per hour × hourly price (p50 is the typical call; the mean, 73 ms, would give \$0.00101); no batching, concurrency or idle time is priced.

Latency run: measured on Colab. CPU Intel(R) Xeon(R) CPU @ 2.00GHz, 8 vCPUs, runtime Colab, Tesla T4 (accelerator unused: CPU benchmark); torch 2.11.0+cu128, onnxruntime 1.30.0. Split val, n = 74, 3 repeats after 10 warm-up calls, batch-1. Shared cloud vCPUs, not an idle-machine run.

H. Reproduction and Determinism

Deliverables. (a) report.pdf (this file). (b) Hugging Face model with label mapping and model card: <https://huggingface.co/gauravgandhi2411/multilingual-intent-router>. (c) W&B report: <https://wandb.ai/gauravgandhi429-gaurav-gandhi/multilingual-intent-router/reports/Submission-summary--VmlldzoXODA1MTA3Mw==>. (d) GitHub repository: <https://github.com/gaurav-gandhi-2411/multilingual-intent-router>; Colab notebook: https://colab.research.google.com/github/gaurav-gandhi-2411/multilingual-intent-router/blob/v1.0-submission/notebooks/intent_router_colab.ipynb.

Reproduce. (a) *Colab*: Runtime > Run all, then upload your own copy of the dataset at the upload cell; **ESTIMATE** on a T4 about 9 to 11 min (training ~6, Track B ~3 to 5; other cells not timed); ends with a REPRODUCTION: PASS/FAIL line against the committed results (outputs in results_colab/). (b) *Locally* (README; PYTHONPATH=src; seed 42): python -m venv .venv && .venv/Scripts/python -m pip install -r requirements.txt; python -m intent_router.final --config configs/final.yaml --stage train; python -m intent_router.final --config configs/final.yaml --stage evaluate; python -m intent_router.trackb --config configs/trackb.yaml --stage headline.

Colab reproduction (**MEASURED**; T4 runtime, notebook run top to bottom, comparison recomputed from the saved outputs). **REPRODUCTION: PASS:** 6 of 6 metrics within tolerance, fresh against committed.

Metric	fresh	committed	delta	tolerance	result
Track A test macro_f1	0.9472	0.9465	+0.0007	±0.03	PASS
Track A test accuracy	0.9459	0.9459	+0.0000	±0.03	PASS
Track B maha_ft AUROC (mean of 3 seeds)	0.8690	0.8705	-0.0015	±0.03	PASS
Track B maha_ft rejection@95 (mean of 3 seeds)	0.2958	0.3417	-0.0458	±0.1	PASS
Track B msp AUROC (mean of 3 seeds)	0.8410	0.8412	-0.0002	±0.03	PASS
Track B msp rejection@95 (mean of 3 seeds)	0.2958	0.1958	+0.1000	±0.1	PASS

Edge case, disclosed. The Track B msp rejection@95 (mean of 3 seeds) delta (+0.1000) sits exactly at its tolerance (±0.1): 71 of 240 unknown rows rejected fresh against 47 of 240 committed, with a threshold set from 62 known calibration rows per seed; the fresh MSP count equals the Mahalanobis one (71 of 240) by arithmetic coincidence, their per-seed counts differ. Not a code bug; per-seed counts are in results/colab/reproduction_run2.json.

Determinism (**MEASURED**). Two independent trainings of the final model (seed 42): identical state-dict hash (yes; 7ca22e16...) and epoch history (yes); validation and test logits and float32 probabilities bitwise equal (yes; max abs difference 0.0e+00 on test logits, 0.0e+00 on test probabilities). A later live W&B re-run of the train stage (9 epochs) reproduced fingerprint 7ca22e16... (match: yes), i.e. byte-identical weights. The first and second Colab runs, on separate fresh runtimes, produced the identical fingerprint 84089f35... and identical metrics (all 6 fresh values equal; the run files are results/colab/reproduction_run1.json and results/colab/reproduction_run2.json), so training is bitwise deterministic within a GPU type; it differs only across GPU types (Tesla T4 against NVIDIA GeForce RTX 3070 Laptop GPU, fingerprints 84089f35... against 7ca22e16...), with all metrics within tolerance.

I. Ablations and Bake-Off Detail

Approach, in detail. The task is single-label, 12-way intent routing on 500 rows with a multilingual tail, plus an open-set requirement, so the system has two parts. First, a closed-set classifier: multilingual-e5-base fine-tuned end to end with the standard classification head (dense + tanh + linear) on the first token (learning rate 3.0e-05, batch 16, max length 64, stopped at epoch 9 of a 20-epoch schedule, seed 42), with temperature scaling fitted on validation rows. Second, a post-hoc open-set score on its penultimate features (Mahalanobis distance to the class means) whose threshold is set on known-class validation rows only, so that 95.0% of known rows are kept; messages below it go to an unknown path instead of a handler. Floors: B0 (character TF-IDF + LR) and B1 (frozen multilingual-e5-base embeddings + LR). Candidates (all AutoModelForSequenceClassification): multilingual-e5-base, xlm-roberta-base, mdeberta-v3-base (learning rate swept per model, then 45 fold-runs for the best). Rule fixed before any result: highest mean CV macro-F1; a smaller or faster model within one std wins; top two compared with the Nadeau-Bengio corrected t-test.

Candidate	LR	CV macro-F1 (mean ± std)	CV acc.	epoch	runs	params (M)	s / run
B0: char TF-IDF + LR		0.785 ± 0.043	0.800	n/a	15	n/a	92.1
B1: frozen e5 + LR		0.914 ± 0.024	0.914	n/a	15	n/a	46.7
F1: multilingual-e5-base (winner)	3e-05	0.939 ± 0.022	0.940	9	45	278	87.6
F2: xlm-roberta-base	5e-05	0.914 ± 0.033	0.918	20	45	278	101.5
F3: mdeberta-v3-base	5e-05	0.900 ± 0.029	0.902	19	45	279	167.6

MEASURED **OPTIMISTIC** CV on train+val only; test untouched. The first rule picks e5 (the others fall below its mean minus one std, 0.917); the second rule does not trigger (best rival is 0.025 behind, inside one std: no). Top-two Nadeau-Bengio test (J = 15): t = 2.26, p = 0.040, so the pre-registered rule, not the p-value, decided.

Why e5-base. It has the highest CV macro-F1 (0.939 ± 0.022) against 0.914 for xlm-roberta-base and 0.900 for mdeberta-v3-base, and no rival is smaller or faster (parameters and seconds per run: table above). Cost of that choice: the fp32 weights are 1112 MB, and on CPU at batch-1 the shipped ONNX fp32 backend has p50 71 ms on one thread and 111 ms with 8 threads (measured on Colab; environment stated in Appendix G). int8 quantization did not meet the pre-registered rule, so fp32 ships (Appendix G); the larger multilingual-e5-large (560 M parameters) needed 10703 MB against a 8 GB card and was not trained. The 5 recipe ablations (class weights, label smoothing, layer-wise LR decay, entity masking, hierarchical loss) show no gain over the CV noise band (0.0028) or in a paired test (tables below). A post-hoc selection check gave the two rivals wider learning-rate and epoch ranges (6 configs): the best reached 0.925 (xlmr_lr7e-05) against e5's 0.944 on the same folds.

Config	CV macro-F1	epoch	runs	VRAM (MB)	Ablation vs e5 baseline	delta CV F1	p	epoch	adopted
e5_lr2e-05	0.939 ± 0.031	14	5	5387	class-weighted CE	-0.0027	0.715	7	no
e5_lr3e-05	0.939 ± 0.022	9	45	5391	label smoothing	-0.0021	0.760	9	no
e5_lr3e-05_class_weight	0.936 ± 0.022	7	15	5388	layer-wise LR decay	-0.0057	0.497	19	no
e5_lr3e-05_entity_mask	0.929 ± 0.034	20	15	5388	entity masking	-0.0092	0.522	20	no
e5_lr3e-05_hier_aux	0.934 ± 0.020	10	15	5388	hierarchical auxiliary loss	-0.0047	0.627	10	no
e5_lr3e-05_label_smoothing	0.937 ± 0.018	9	15	5388	MEASURED OPTIMISTIC Each had to beat the CV noise band (0.0028) and pass a paired Nadeau-Bengio test at the pre-registered level; 15 baseline runs.				
e5_lr3e-05_ild	0.933 ± 0.023	19	15	5061					
e5_lr5e-05	0.936 ± 0.025	7	5	5388					
mdeberta_lr2e-05	0.867 ± 0.037	18	5	5424					
mdeberta_lr3e-05	0.898 ± 0.043	16	5	5424					
mdeberta_lr5e-05	0.900 ± 0.029	19	45	5425					
xlmr_lr2e-05	0.919 ± 0.022	17	5	5385					
xlmr_lr3e-05	0.919 ± 0.016	13	5	5385					
xlmr_lr5e-05	0.914 ± 0.033	20	45	5388					

MEASURED **OPTIMISTIC** Configs with few runs were learning-rate sweep points (one fold seed, one model seed); only the best rate per model got the full confirm stage.

Track B Improvement Rounds: Scorer Comparison, Robustness Variants and Selection Axes

Candidate scorer	DEV AUROC
current: Mahalanobis (fine-tuned), shipped	0.894
C1: rank fusion Mahalanobis + energy	0.890
C1: multi-layer Mahalanobis	0.832
C2: CE + SupCon, lambda low	0.887
C2: CE + SupCon, lambda high	0.884
C4: 5-seed ensemble (evidence only)	0.896
B1 fairness variant: frozen e5 + LR, MSP	0.836

MEASURED Selection on DEV only. Best eligible candidate minus the current method -0.0031 against a required margin of 0.0115; ship: no. C2 passed the Track A guard (0.9401, 0.9447 vs 0.9341) but lowered DEV AUROC. C4 (evidence only): CONFIRM 0.893 vs 0.879; B1 fairness variant on CONFIRM 0.856. C3: 560 M parameters, static fp32 weights, grads and Adam state 8543 MB before activations; aborted: yes.

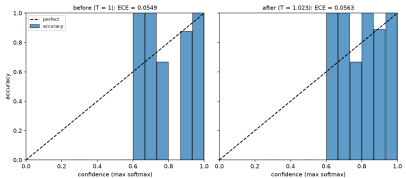
Robustness and open-set training variants. A1 (ID-prefix randomisation) cut neutral-swap flips (reduction +0.250 [+0.155, +0.345]) but did not meet the Track A guard (0.9261 vs 0.9341); A2 (outlier exposure) did not meet the guard (0.9214) and lowered DEV AUROC (-0.037 [-0.056, -0.019]) and rejection (-0.126 [-0.198, -0.056]); A3 passed (0.9562) with zh agreement gain +0.050 [+0.011, +0.092] and mean-language gain +0.048 [+0.027, +0.071], and was adopted as v2. Multi-axis selection axes are candidate minus v1 as improvement (positive is better; CI of the improvement); rejection is strict rejection at the high retention operating point.

Candidate	epoch	(a) CV F1	(b) DEV AUROC	(b) DEV rejection	(c) flip rate	(d) transl. agreement	(e) ECE	eligible	improved axes
A3 (MT + noise copies)	6	+0.013 [+0.005, +0.022]	+0.000 [-0.012, +0.013]	+0.002 [-0.047, +0.050]	-0.127 [-0.214, -0.044]	+0.056 [+0.040, +0.072]	+0.001 [-0.007, +0.013]	no	2
A1 (ID-prefix randomisation)	11	-0.011 [-0.021, -0.001]	-0.014 [-0.028, -0.000]	-0.053 [-0.087, -0.019]	+0.202 [+0.139, +0.270]	-0.020 [-0.036, -0.005]	-0.011 [-0.024, +0.000]	no	1
A1 + A3 (v3)	6	+0.000 [-0.010, +0.011]	-0.002 [-0.018, +0.013]	-0.019 [-0.067, +0.031]	+0.187 [+0.123, +0.250]	+0.046 [+0.027, +0.066]	+0.000 [-0.011, +0.012]	yes	2

MEASURED v1 reference at epoch 9: CV F1 0.934, DEV AUROC 0.880, rejection 0.420, flip rate 0.238, agreement 0.853, ECE 0.026. Seeds averaged inside each paired-bootstrap resample; no multiplicity correction across candidates and axes, which is why eligibility, not only improvement, gates selection.

A2 Track A detail: calibration, per-class out-of-fold scores, slices and error analysis

Paired comparison on the same test rows. Fine-tuned minus B0: +0.213 [+0.113, +0.340] (McNemar exact: 17 vs 2 discordant, p = 7.3e-04); minus B1: +0.048 [-0.009, +0.119] (5 vs 1, p = 0.219). Parent-level accuracy 0.946. **Calibration.** Temperature 1.023; test ECE 0.055 before and 0.056 after scaling (no gain on test; validation, in-sample: 0.083 to 0.068). Selective prediction at the shipped threshold: coverage 0.986, accuracy on accepted 0.959, AUROC 0.0051.



MEASURED reliability, test, before and after temperature scaling.

Class	test, single evaluation				out-of-fold OPTIMISTIC			
	P	R	F1	n	P	R	F1	n
ai_agent_performance	1.000	1.000	1.000	6	1.000	1.000	1.000	34
appointment_manager	1.000	0.857	0.923	7	0.925	0.974	0.949	38
chitchat	1.000	1.000	1.000	5	1.000	1.000	1.000	30
customer_support	0.800	0.667	0.727	6	0.968	0.882	0.923	34
document_processing	0.833	0.833	0.833	6	0.935	0.853	0.892	34
knowledge_base	1.000	1.000	1.000	6	0.861	0.912	0.886	34
orders	0.875	1.000	0.933	7	0.947	0.947	0.947	38
other	1.000	1.000	1.000	4	1.000	0.923	0.960	26
shipment_information.analytics	1.000	1.000	1.000	8	0.939	0.979	0.958	47
shipment_information.disruptions	1.000	1.000	1.000	5	1.000	1.000	1.000	30
shipment_information.realtime_query	0.889	1.000	0.941	8	1.000	0.957	0.978	47
yard_management	1.000	1.000	1.000	6	0.895	1.000	0.944	34

MEASURED Out-of-fold: 426 train+val rows, probabilities averaged over 9 predictions per row, then scored once with the test definition (computed from the saved OOF predictions; macro-F1 equals the stored 0.953 [0.932, 0.971]). Post-selection, so optimistic.

Slice (n_chars < 30; indicative if n < 20)	out-of-fold OPTIMISTIC				test, single evaluation	
	n	macro-F1 [CI]	acc. [CI]	n	macro-F1 [CI]	acc. [CI]
all rows	426	0.953 [0.932, 0.971]	0.953 [0.932, 0.972]	74	0.947 [0.865, 0.989]	0.946 [0.892, 0.986]
English	370	0.949 [0.924, 0.969]	0.949 [0.924, 0.970]	60	0.940 [0.825, 0.988]	0.933 [0.867, 0.983]
non-English (primary language)	49	0.963 [0.821, 1.000]	0.980 [0.939, 1.000]	13*	1.000 [0.500, 1.000]	1.000 [1.000, 1.000]
Spanish	17*	1.000 [0.600, 1.000]	1.000 [1.000, 1.000]	5*	1.000 [0.500, 1.000]	1.000 [1.000, 1.000]
French	14*	0.900 [0.500, 0.900]	0.929 [0.786, 1.000]	n/a		
German	10*	1.000†	1.000 [1.000, 1.000]	n/a		
Chinese	6*	1.000 [0.400, 1.000]	1.000 [1.000, 1.000]	n/a		
language undetermined	7*	1.000 [0.667, 1.000]	1.000 [1.000, 1.000]	1*	n/a	1.000 [1.000, 1.000]
code-mixed	1*	n/a	1.000 [1.000, 1.000]	0*	n/a	n/a
short texts	29	1.000 [0.667, 1.000]	1.000 [1.000, 1.000]	3*	1.000 [0.500, 1.000]	1.000 [1.000, 1.000]
noisy	223	0.958 [0.928, 0.981]	0.955 [0.928, 0.982]	41	0.956 [0.733, 1.000]	0.951 [0.878, 1.000]
clean	203	0.943 [0.889, 0.974]	0.951 [0.921, 0.980]	33	0.875 [0.607, 0.909]	0.939 [0.848, 1.000]
shipment_information.* family	124	0.989 [0.974, 1.000]	0.976 [0.944, 1.000]	21	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]

MEASURED * = indicative only (small n); shaded rows contain such a slice. Slice macro-F1 uses the slice's gold classes, fixed across resamples, so intervals are biased low when a resample drops a class; † = the point lies outside its own interval for that reason, so the interval is not shown. Single-row slices report accuracy only.

Error analysis detail. Hypothesis: errors concentrate in shipment information.* (rejected). MEASURED OPTIMISTIC Out-of-fold, 45 fold-runs (3834 predictions): the family is 29.1% of rows and 16.9% of errors; the cluster-bootstrap interval of (error share minus row share) is [-0.224, -0.004] (10000 resamples over row ids), so the family is under-represented in errors. Of its 39 family errors 15 stay inside the family (38.5%). On test: 4 errors, 0 in the family.

Gold → predicted (45 fold-runs)	count
document_processing → knowledge_base	27
knowledge_base → document_processing	18
customer_support → appointment_manager	14
other → knowledge_base	10
customer_support → yard_management	9

MEASURED OPTIMISTIC Hypotheses (next step: targeted test): document_processing vs knowledge_base share document vocabulary (lexical overlap); customer_support catches complaints that mention scheduling or yard events (missing context). Test split: appointment_manager → orders; customer_support → document_processing; customer_support → shipment_information.realtime_query; document_processing → customer_support.

J. Test-Split Exposure

Test-split exposure. Each final closed-set model (v1, v2, v3) was test-evaluated exactly once, and so were the baselines B0 and B1; every other call that read the test split is inference-only (robustness, LLM baseline, quantization, latency, determinism, Hub round trip) and computes no Track A metric, with the counts taken from the test-evaluation record files below. The leave-one-class-out and Track B selection models (never the final model) read known-class test rows as the known side of the open-set evaluation (Open-set improvement round added more such reads, counted below, and no classifier test evaluation), so the v1-versus-v3 decision, which used the Track B headline and CONFIRM holdouts, was not fully test-free (the selection rules were fixed before the candidates were run). The Track A test macro-F1 of the final models played no part in any selection. Reproduction runs (the notebook, locally and on Colab) re-evaluate the bit-identical v1 weights into results_colab/, recorded separately from results/; they feed no selection and no reported Track A number.

Inference-only call type in the test-evaluation record	lines	Track B / selection records (LOCO and DEV models, known-class test rows)	lines
robustness (ID swap, noise, translation), no Track A metric	66	trackb / test_inference_log.jsonl	16
LLM baseline inference	8	trackb_improve / test_inference_log.jsonl	76
int8 / ONNX quantization comparison	6	phase4c / test_inference_log.jsonl	23
latency measurement	5	phase4e / test_inference_log.jsonl	60
bit-for-bit determinism re-run	3	phase4e / confirm_test_inference_log.jsonl	8
Hub round-trip reproduction check	2	phase6a / test_inference_log.jsonl	75
		phase6a / diag / test_inference_log.jsonl	48
		phase6a / confirm_test_inference_log.jsonl	8

Final and baseline models: evaluation lines	fingerprint	date	git sha
v1	7ca22e16	2026-10-02	696df48
v2	db5b5808	2026-10-02	39b342a
v3	78df8842	2026-10-02	06d5d20
B0	bbe5b7ab	2026-10-02	696df48
B1	4daba63f	2026-10-02	696df48

MEASURED Counted by code from the record files. results/final/test_eval_log.jsonl holds 95 lines: 3 evaluation lines for 3 distinct final fingerprints, 2 for the B0 and B1 baselines and 90 inference-only lines. The Track B records hold 183 lines across 5 files. Open-set improvement round added 131 Track B test-row reads across 3 further record files (rows above, by call type; unknown-side held-out-class rows included) and made no new classifier test evaluation.

K. Further Detail

Headline summary, full

Headline	Value	What it is
Track A: test macro-F1 (primary)	0.947 [0.865, 0.989]	MEASURED one recorded test evaluation, n = 74, model seed 42, 95% percentile bootstrap (10000 resamples, seed 42). Shipped model: v1.
Track A: test accuracy	0.946 [0.892, 0.986]	MEASURED same evaluation
Track A: cross-validated macro-F1	0.939 ± 0.022	MEASURED OPTIMISTIC 45 fold-runs (3 fold seeds × 3 model seeds × 5 folds); epoch and model were chosen on these folds
Track A: out-of-fold macro-F1	0.953 [0.932, 0.971]	MEASURED OPTIMISTIC 426 train+val rows, probabilities averaged over 9 predictions per row
Floors on the same test split (B0, B1)	0.733 [0.602, 0.822] 0.899 [0.802, 0.956]	MEASURED B0 above (char TF-IDF + LR), B1 below (frozen e5 + LR); same test split
Track B: strict rejection recall at the 95% retention operating point	0.342 [0.263, 0.425]	MEASURED headline holdout (document_processing + yard_management), 3 model seeds, rows resampled within the holdout, 10000 resamples. Measured limit: see section 4.
Track B: known-class retention / AUROC	0.968 [0.925, 1.000] 0.870 [0.811, 0.923]	MEASURED same runs (retention above, AUROC below)

Data quirks, full list

- Fewer non-English rows than stated.** Language ID (MEASURED lingua, offline) finds 12.4% non-English by primary language (code-mixed 0.2%), not the stated ~25.0%; 8 rows are undetermined and never counted. Spanish 22, French 18, German 12, Chinese 7 rows: small slices, so each carries n and an interval.
- Short and noisy text.** 6 rows are very short (emoji, one-liners): kept. 52.8% of rows are noisy (33.7% all-lower-case, 6.8% slang). Token p99 is 36.0; max length 64.
- Near-duplicates.** 2 pairs above character-TF-IDF cosine 0.8, same label, are grouped so they never straddle splits or folds; cross-label pairs above 0.7: 0.
- Class imbalance.** Ratio 1.83: macro-F1 and stratified, duplicate-grouped splits; class-weighted cross-entropy was ablated, not adopted (CV delta -0.0027, p = 0.715).
- Catch-all classes.** chitchat (35 rows) and other (30 rows) are the smallest classes (test F1 1.000, n = 5; 1.000, n = 4: small n, indicative). In Track B lenient mode they count as safe landings, but 0 of 158 un-rejected unknowns land there, so lenient equals strict.
- Small data.** Weight decay 0.01 and the stopping epoch (9) were chosen on CV, never on test; headline numbers carry a CV or bootstrap interval.
- ID-prefix shortcut.** PO-style IDs occur in 23 rows, 82.6% labelled orders (flagged: yes); LD-style IDs in 44 rows, 72.7% shipment_information.realtime_query. A masked-entity ablation reduced CV F1 (-0.0092); the counterfactual damage is in Appendix E.

- **Sibling overlap and label ambiguity.** The three shipment classes overlap lexically (nearest-neighbour accuracy 0.594). An LLM label-ambiguity audit ([MEASURED](#), 50 items, Fleiss kappa 0.685) did not meet its pre-registered calibration gate (judge accuracy 0.70 / 0.70 / 0.80 vs 0.90), so its labels are not used and nothing is relabelled. Of 20 model errors the judges' consensus sided with gold in 9, the model in 4, a third label in 4, none in 3.

Evaluation protocol

Why macro-F1, CV and CIs. Classes are equal-weight for routing, and the test split is 74 rows (about 4 to 8 per class), so one error moves a class F1 a lot; we pair the split with CV and intervals. **Splits.** Stratified and grouped by near-duplicate group: train 352, validation 74, test 74; CV runs on the remaining 426 rows. **Leakage controls.** Temperature and the abstention threshold are fitted on validation only; Track B thresholds use known-class calibration rows only; the audit of the 16 Track B runs found all assertions passing: yes (zero held-out-class rows in any training set).

What to monitor and how drift is caught

- **Input and output drift:** abstention rate (test reference: 0.986 coverage), confidence and negative-energy distributions, language mix against the measured 12.4% non-English, kNN distance to train compared with a reference window (PSI or MMD), per-intent volume, downstream handler failure rate.
- **Label drift:** weekly sampled audits labelled by consensus of different model families, blind, with agreement measured and the result labelled consensus, not ground truth. Our pilot (50 items) had Fleiss kappa 0.685 but did not meet calibration (reliable: no), so humans label any sample that triggers a retrain.
- **13th-class playbook:** cluster the abstained traffic, name and label the cluster, add the class immediately through the kNN bank (no retraining), then re-train fully, shadow-evaluate against v1, and promote. Retraining triggers: sustained rise in abstention rate, kNN-distance PSI above its alert level, a handler failure spike, or a new labelled cluster of meaningful size.

Other limitations

v2 and v3 lowered strict rejection on the scored Track B holdout and are not shipped; the LLM label audit did not meet its calibration gate and is not used. Small data: a 74-row test split, language slices as small as 7 rows (Chinese), MT-evaluated translation rows. **ID-prefix shortcut:** the neutral prefix REF still flips 23.7% of the 38 LD-prefixed OOF rows (9 flips; 0.238 pooled over the neutral swaps in the multi-axis selection axes), so part of the closed-set accuracy rides on ID text. CPU latency was measured on Colab shared vCPUs (environment stated), not on an idle machine of our own.

The 3–4-hour path. The core pipeline (EDA, a stratified split, an e5 fine-tune, macro-F1, an MSP abstention threshold set on held-out data, W&B tracking, the Hub push) produces the Track A result: test macro-F1 0.947 [0.865, 0.989].

The additional effort bought evidence rather than score: confidence intervals, pre-registered selection rules, and the Track B diagnosis (D1 to D4) with its fix plan.

That evidence changes what can be claimed: the gap to B1 (+0.048 [-0.009, +0.119]) sits inside its interval at n = 74 rows and 65.8% of unseen intents are still routed to a known label, neither of which a point estimate alone shows.

Next steps, detail

(a) **N1, verified hard-negative out-of-scope data (the first step).** D1 is the evidence: held-out intents are separable (oracle AUROC 0.969 [0.942, 0.990]) but the unsupervised score reaches 0.870, a missing negative training signal. Generate out-of-scope messages that reuse keywords of the 12 intents, keep those all three judges unanimously label as fitting none, then compare a K-plus-one class head against outlier exposure with a uniform target on the same Track B holdouts. Generic synthetic outlier exposure (A2) did not meet the Track A guard (0.9214 vs 0.9341) and lowered DEV AUROC (-0.037 [-0.056, -0.019]); the judges did not meet calibration (reliable: no) and need a gate first. Published outlier-exposure results are outside these measurements. (b) **More labelled data (D4):** the measured AUROC slope is +0.056 [+0.049, +0.061] per doubling, but the rejection slope (+0.061 [-0.010, +0.145]) has an interval reaching zero. (c) **I4, ID-consistency loss, with its measured trade-off:** neutral-swap flip rate -0.183 and -0.198 (two loss weights; negative is fewer flips) against CV macro-F1 -0.023 [-0.033, -0.013] and -0.023 [-0.032, -0.015], ineligible on the F1 axis; it needs a variant that recovers that loss. v3 cut neutral-swap flips by +0.187 [+0.123, +0.250] but moved headline rejection by -0.133 [-0.204, -0.067], so any such attempt must be guarded on Track B too. (d) **N2, e5-large fine-tune: not worth it under D1 (cut).** On the all-rows pool, the pre-registered rule's metric (frozen encoders trained on none of these rows), frozen e5-large oracle AUROC is 0.9744 against 0.9699 for e5-base, a gap of +0.0045, short of the 0.02 the rule needs (run N2: no). (e) **Closed in the open-set improvement round.** I1: i1a changed raw DEV rejection by -0.077 (ineligible); i1b was selected, then held back by the pre-ship check (headline strict rejection -0.033 [-0.062, -0.008]). I6: i6a retains 0.878 and i6b 0.456 of known DEV rows (collapsed).